

Erdős Problem #208

EP-208

L1 - TRACTABLE

OPEN

Combinatorics

ORIGINAL PROBLEM STATEMENT

Let $s_1 < s_2 < \dots$ be the sequence of squarefree numbers. Is it true that, for any $\epsilon > 0$ and large n , $s_{n+1} - s_n \ll_\epsilon s_n^\epsilon$? Is it true that $s_{n+1} - s_n \leq (1 + o(1)) \frac{\pi^2}{6} \frac{\log s_n}{\log \log s_n}$?

Source and attribution. Statement reproduced from UnsolvedMath, whose official dataset is released under the CC BY 4.0 license.

Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsignedMath>